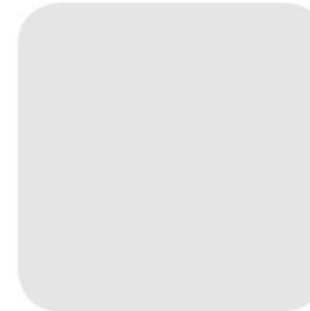
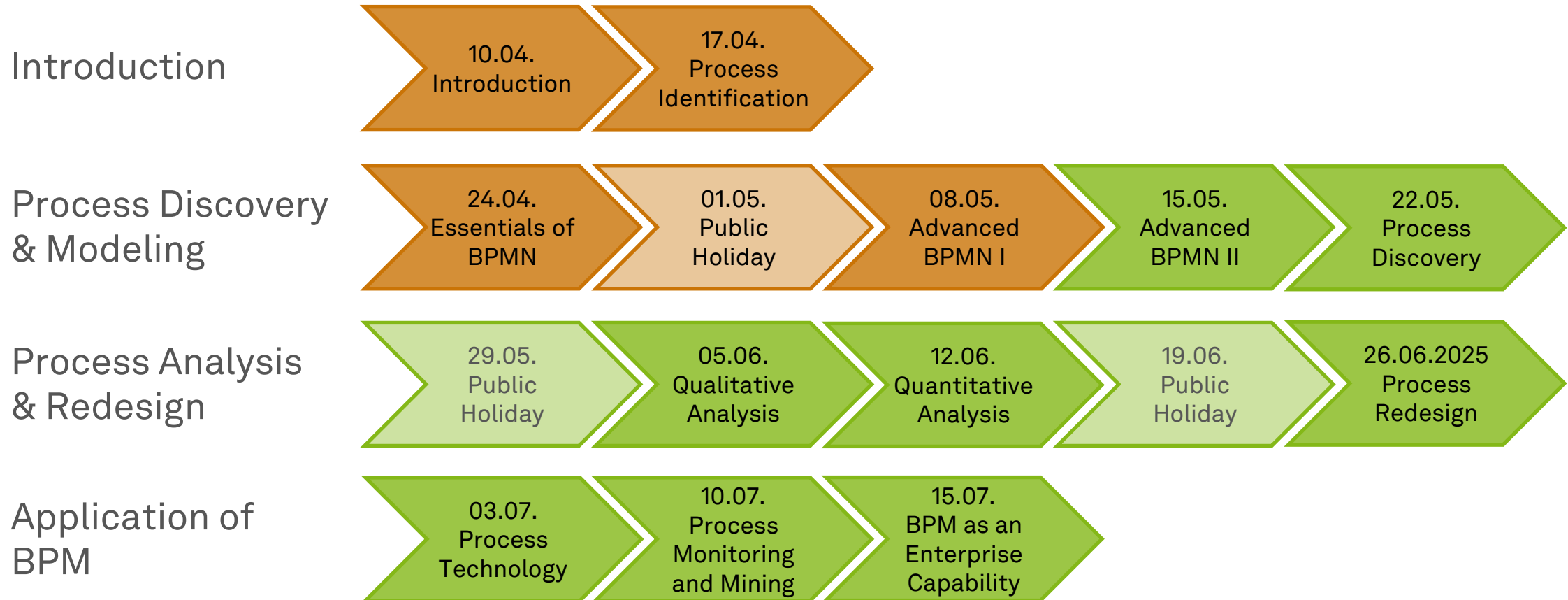


Business Process Management

Advanced Process Modeling II



Roadmap



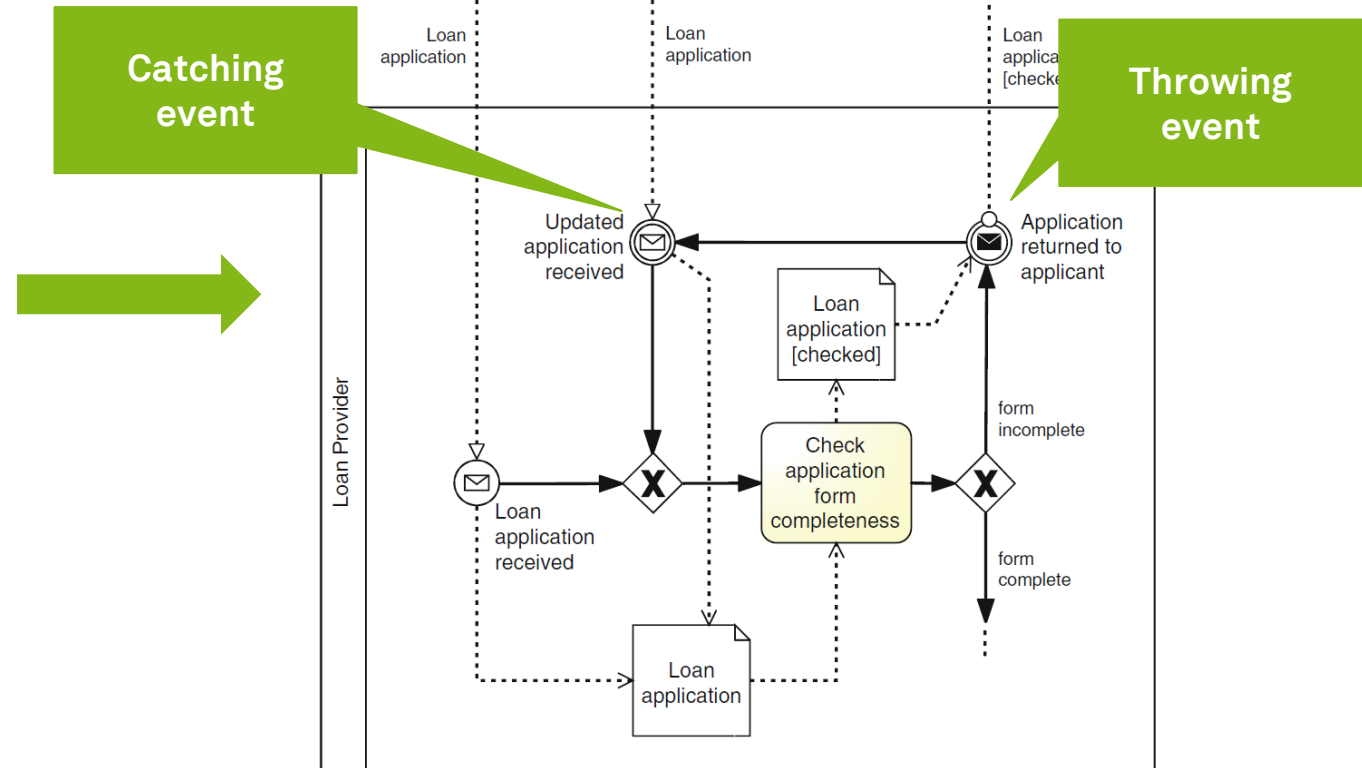
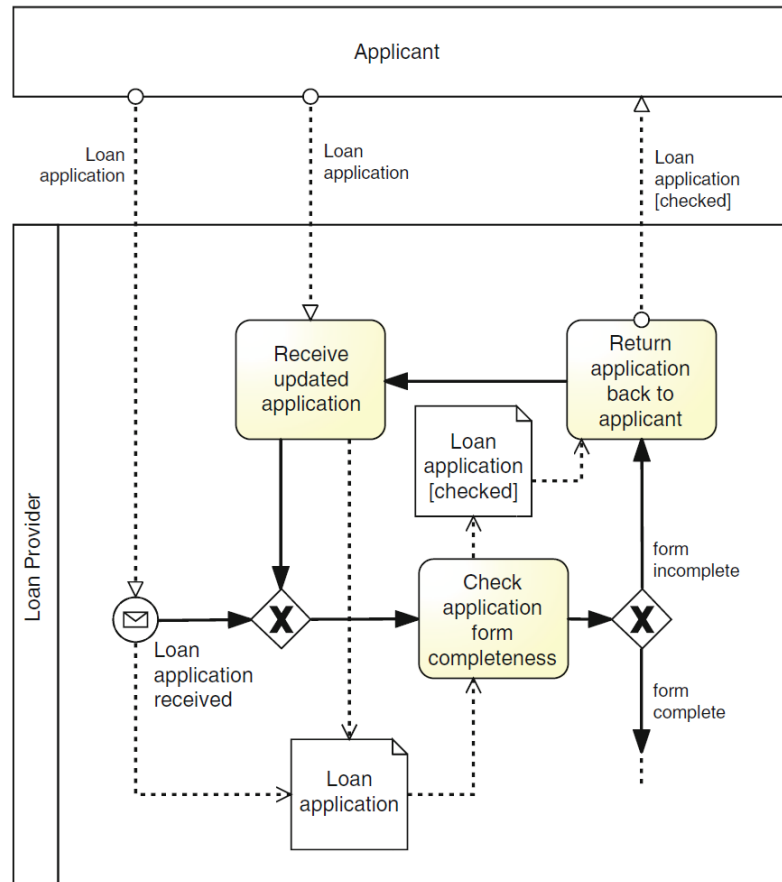
Learning Goals of Today

- Understand throwing and catching events in BPMN
- Understand exception handling in BPMN
- Understand and know workflow patterns
- Be able to use basically all of BPMN

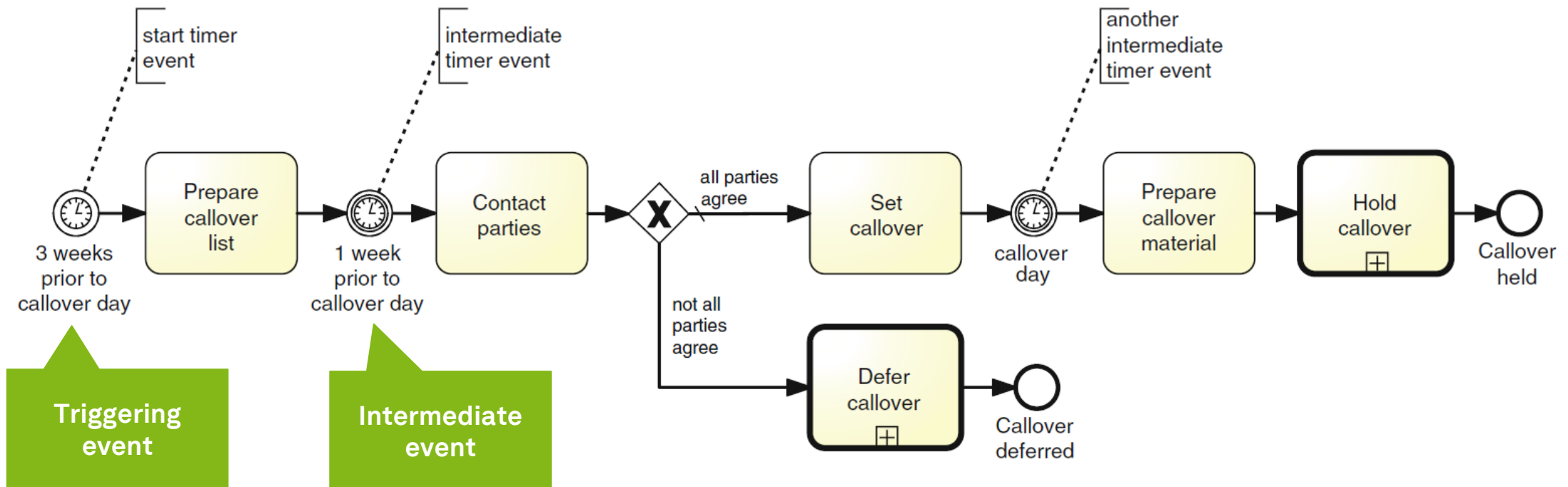
Event Handling



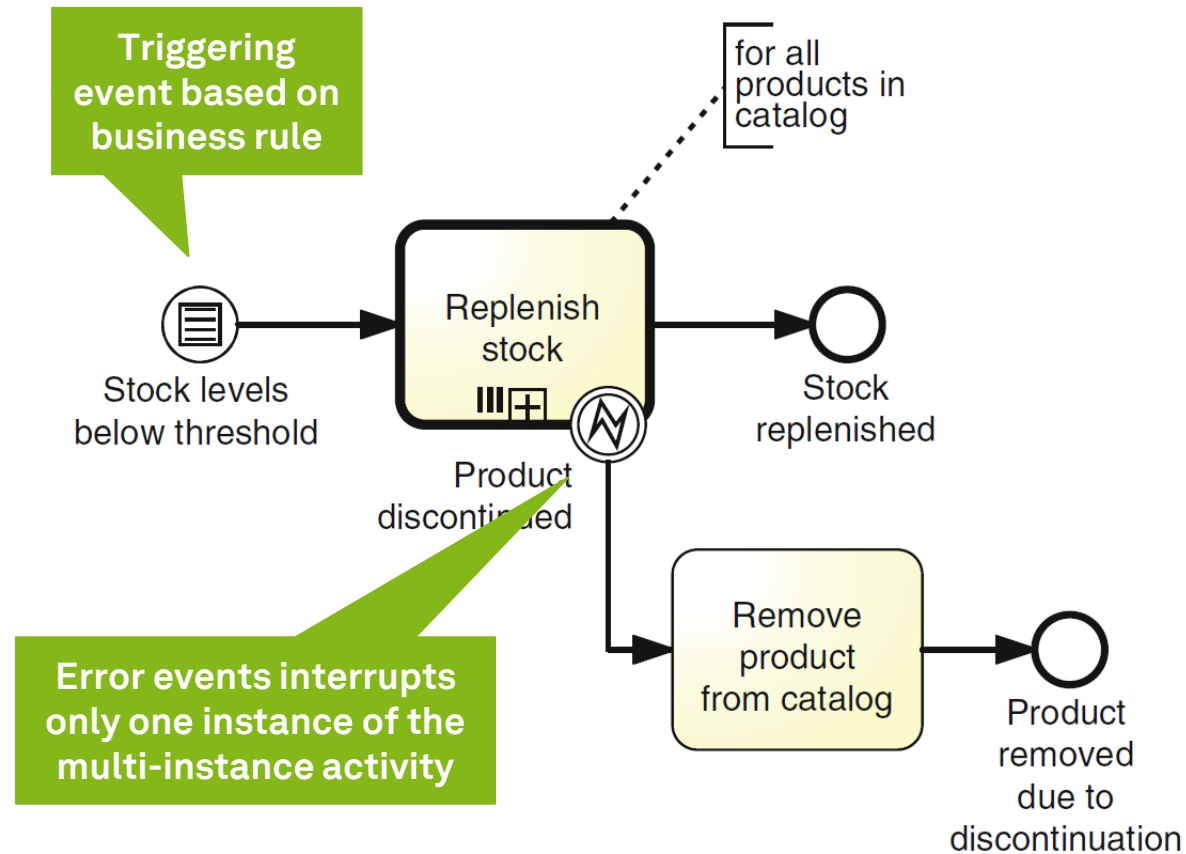
Message Events



Temporal Events



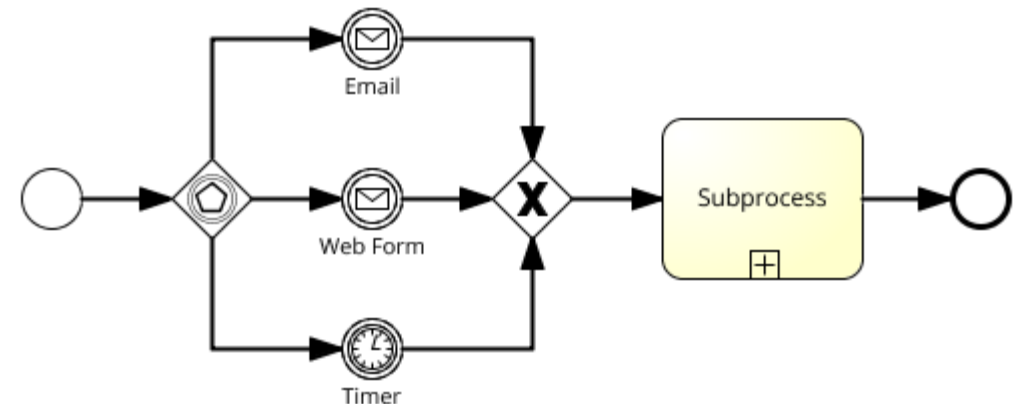
Conditional Events



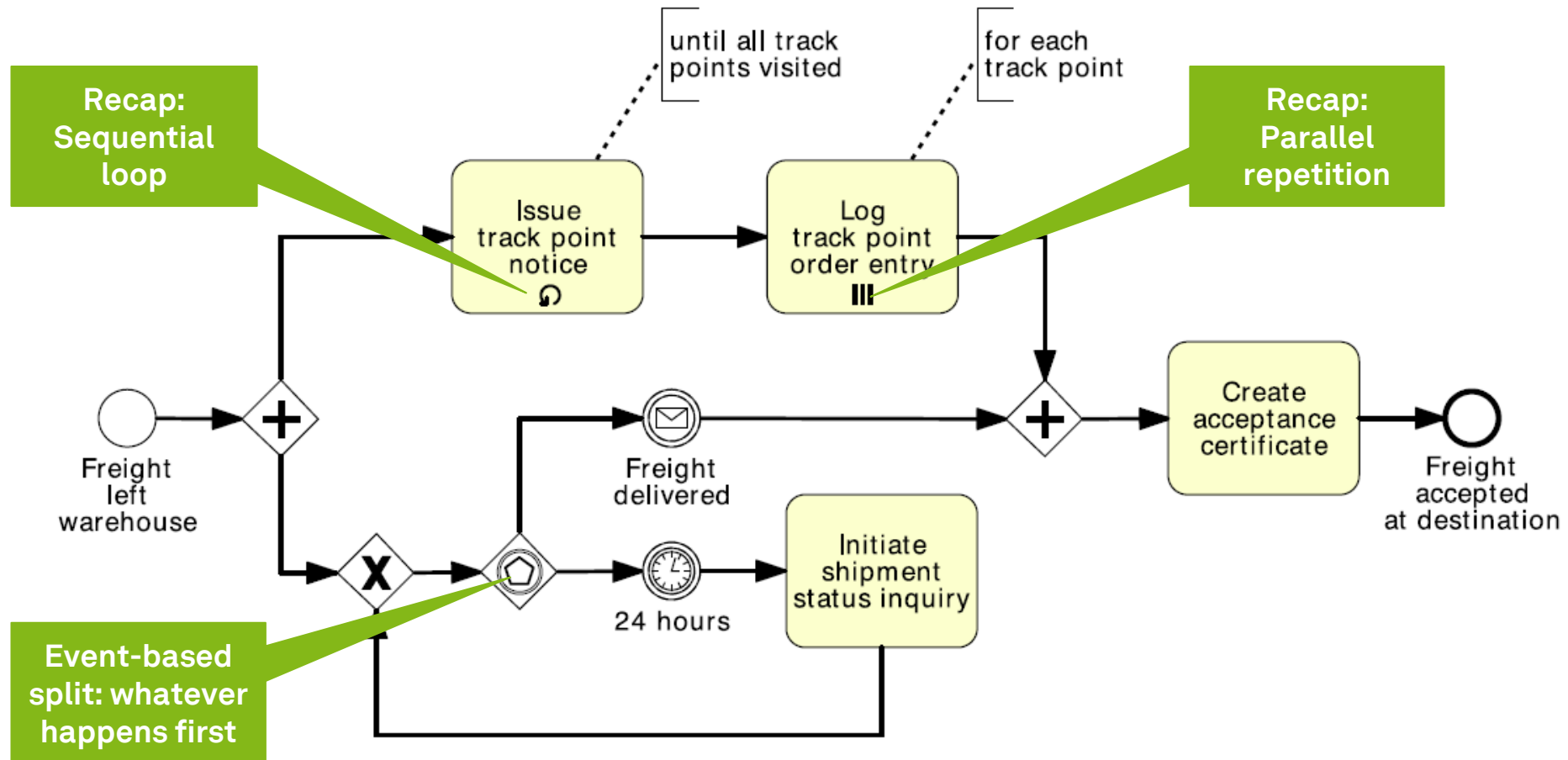
Data-based vs. Event-based XOR-Split

- A branch in an XOR gateway is chosen based on expressions evaluated over available data
- The decision is made **without any delay** (not time consuming)
- However, there are cases in which the decision must be **delayed** until something happens
 - Approval of management
 - Customer decision
 - ...

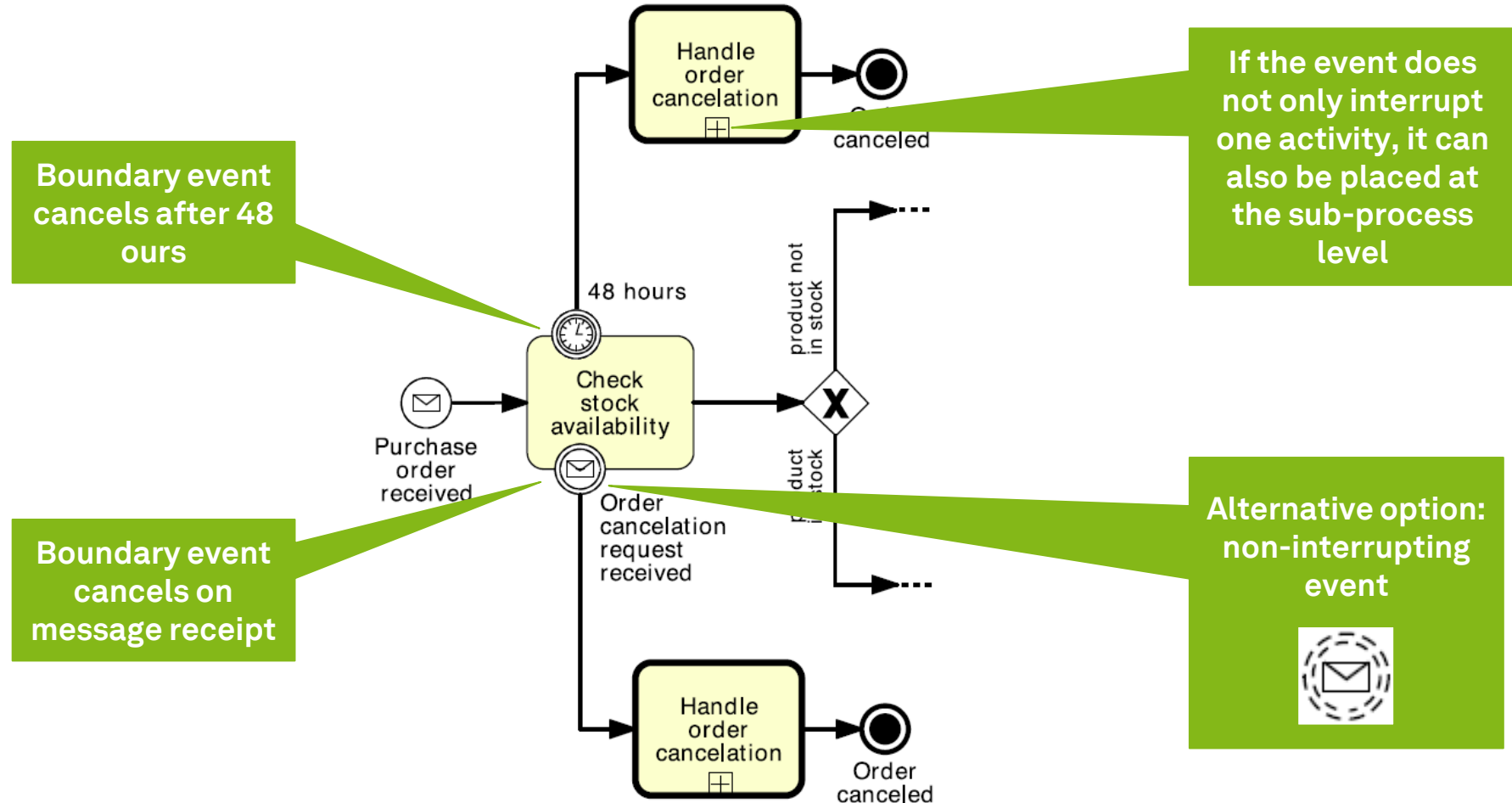
- In this case, BPMN provides an **event-based decision gateway**



Racing Events



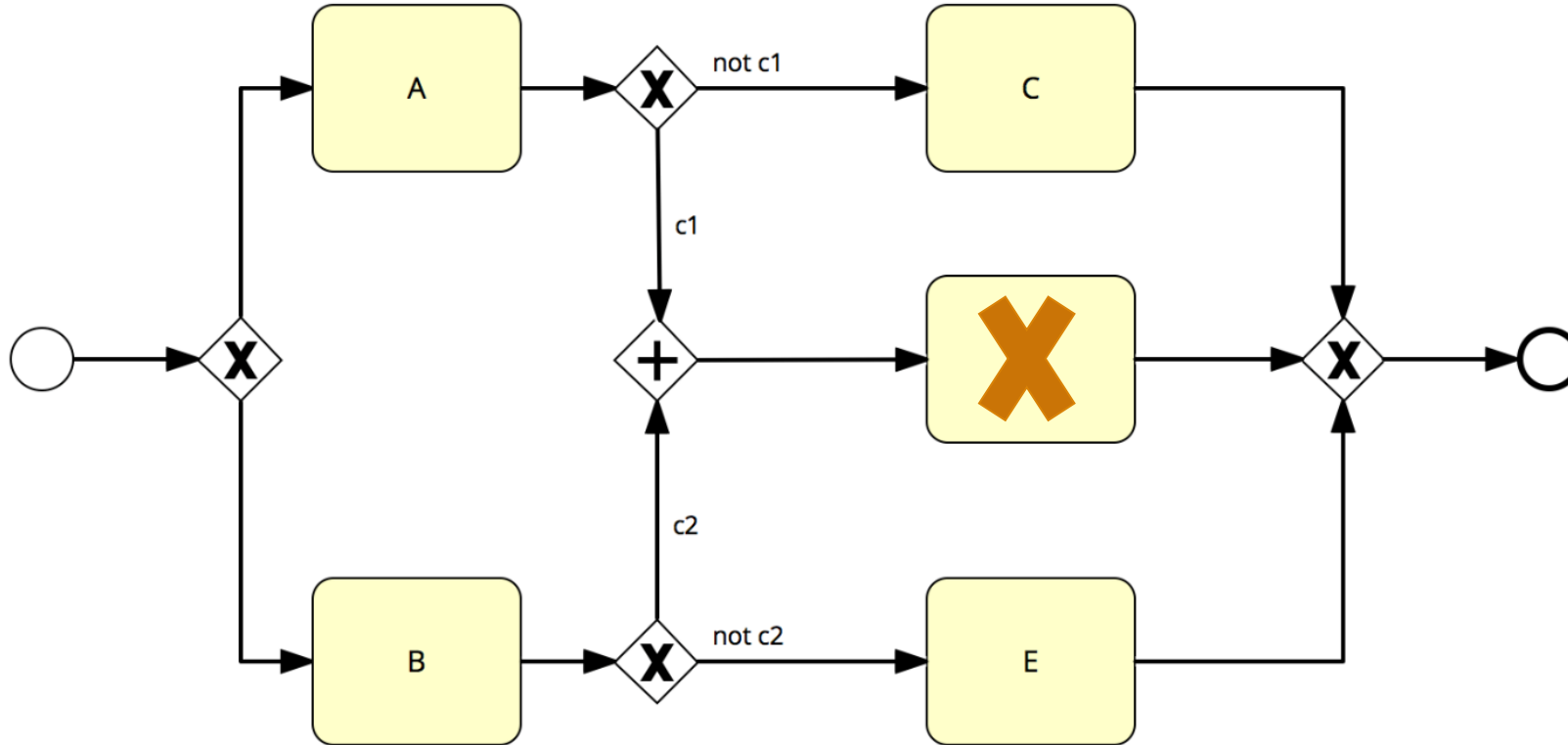
Boundary Events



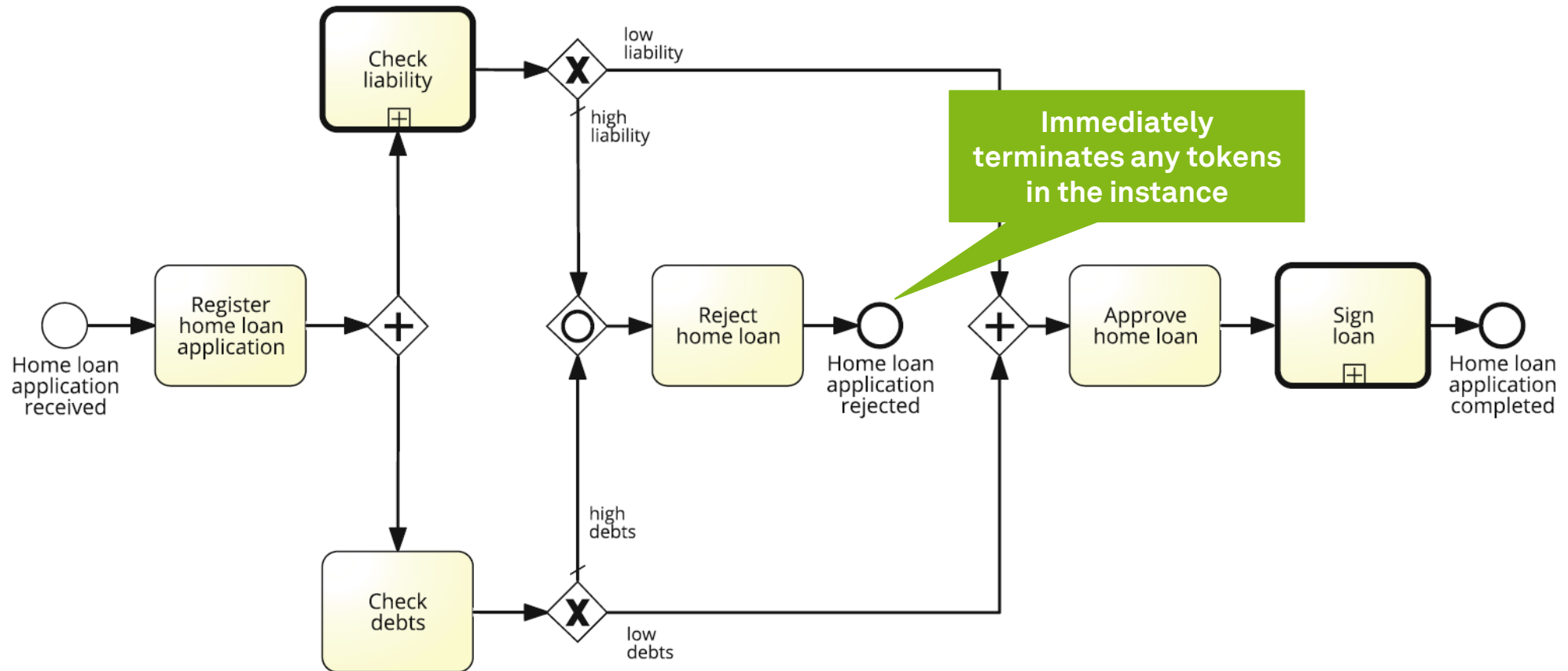
Exception Handling



Instance Termination: Correct or Incorrect?

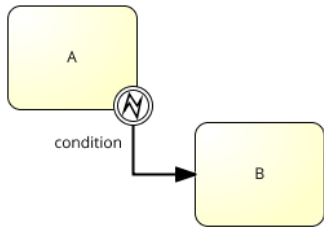


Process Termination



Exception Handling

- Exception handling often means stopping a sub-process and performing a special activity

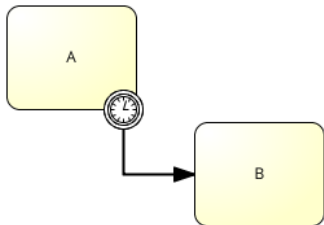
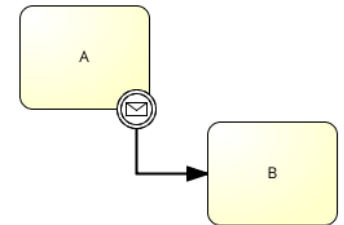


Internal Exception

An error **inside** the activity occurred. The execution of the activity must be interrupted. Handled with **Error** event.

External Exception

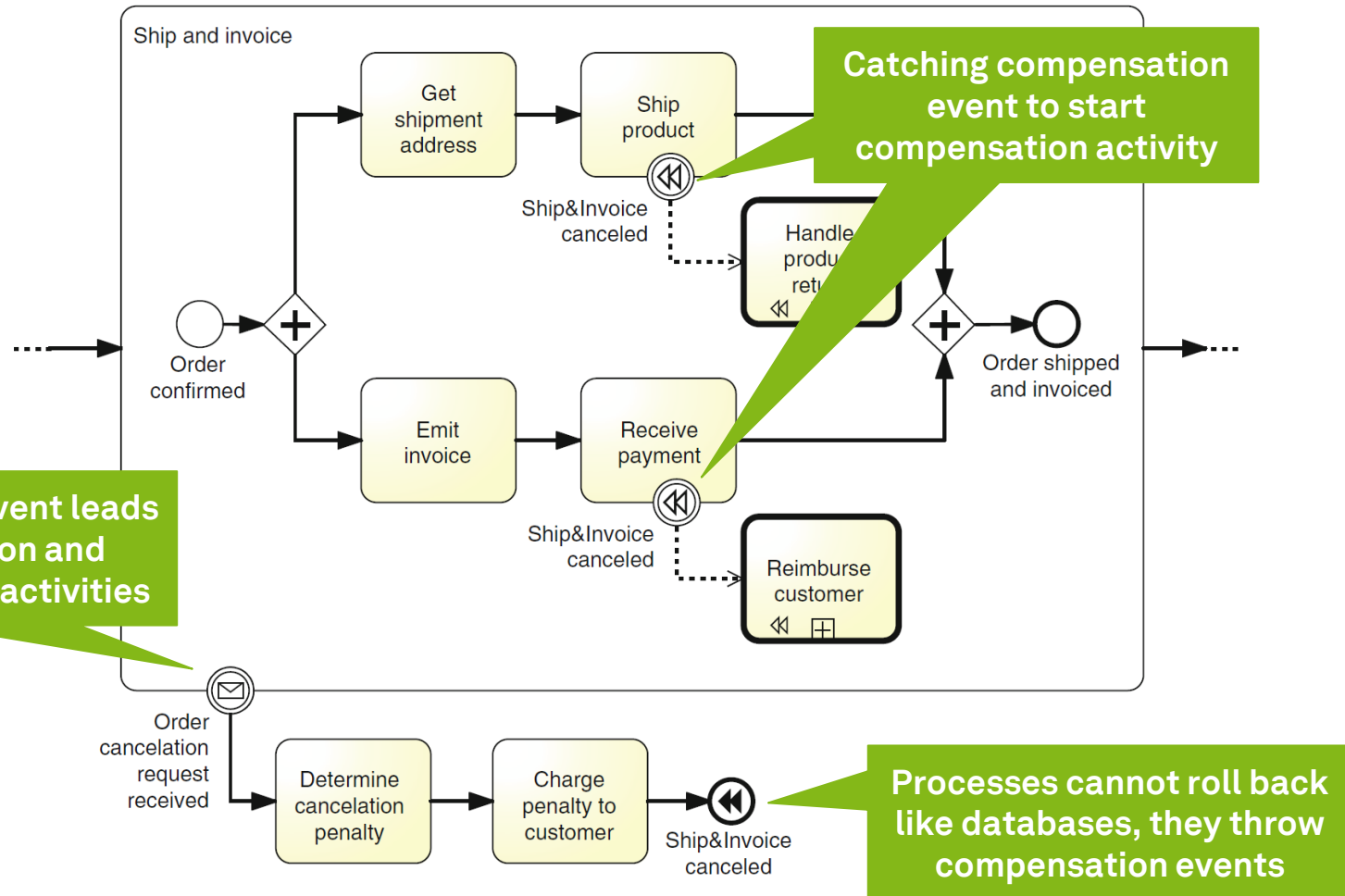
An error **outside** the activity occurred. The execution of the activity must be interrupted. Handled with **Message** event.



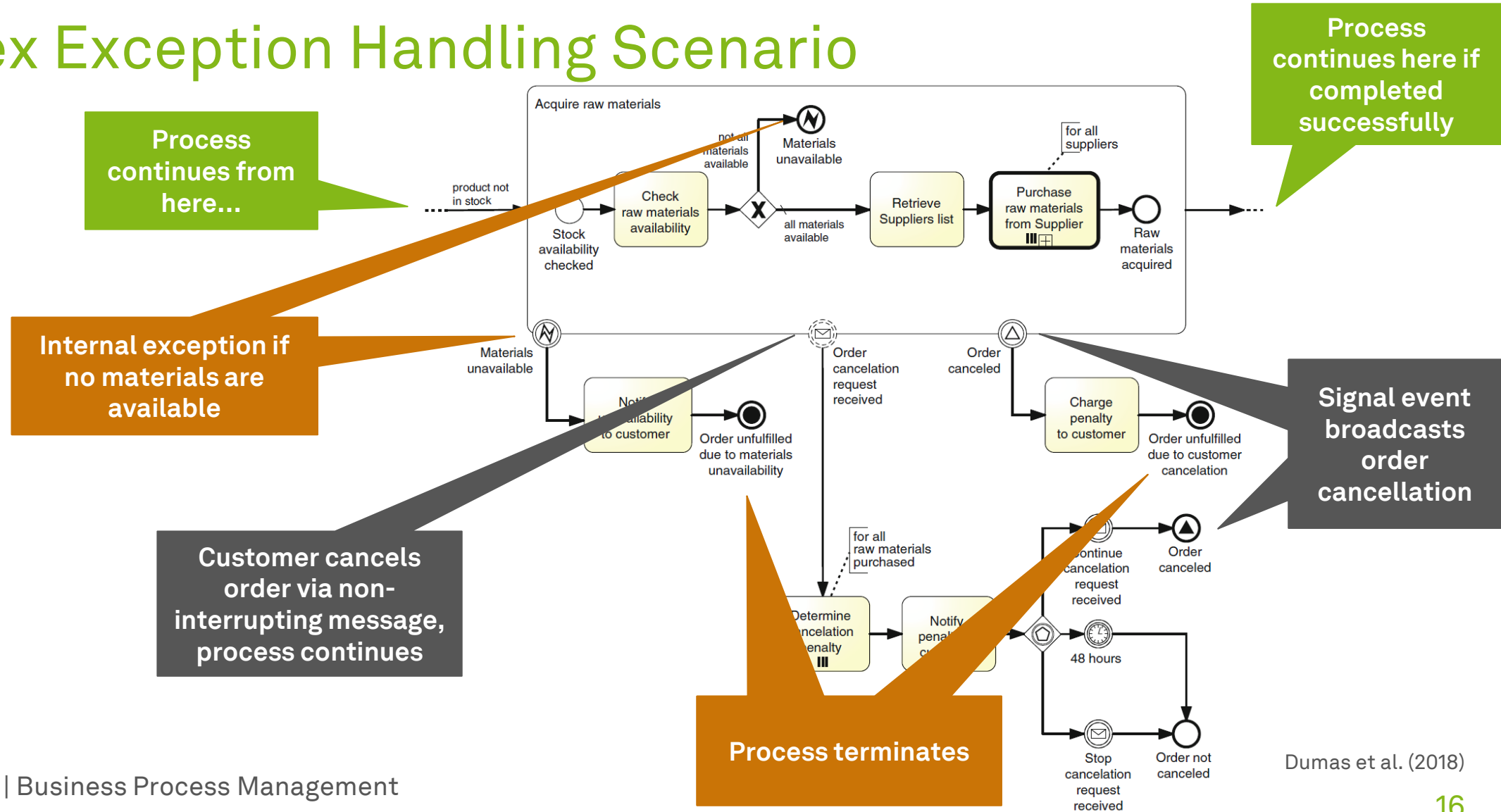
Timeout Exception

An activity lasts **too long** and must be interrupted. Handled with the **Timer** event.

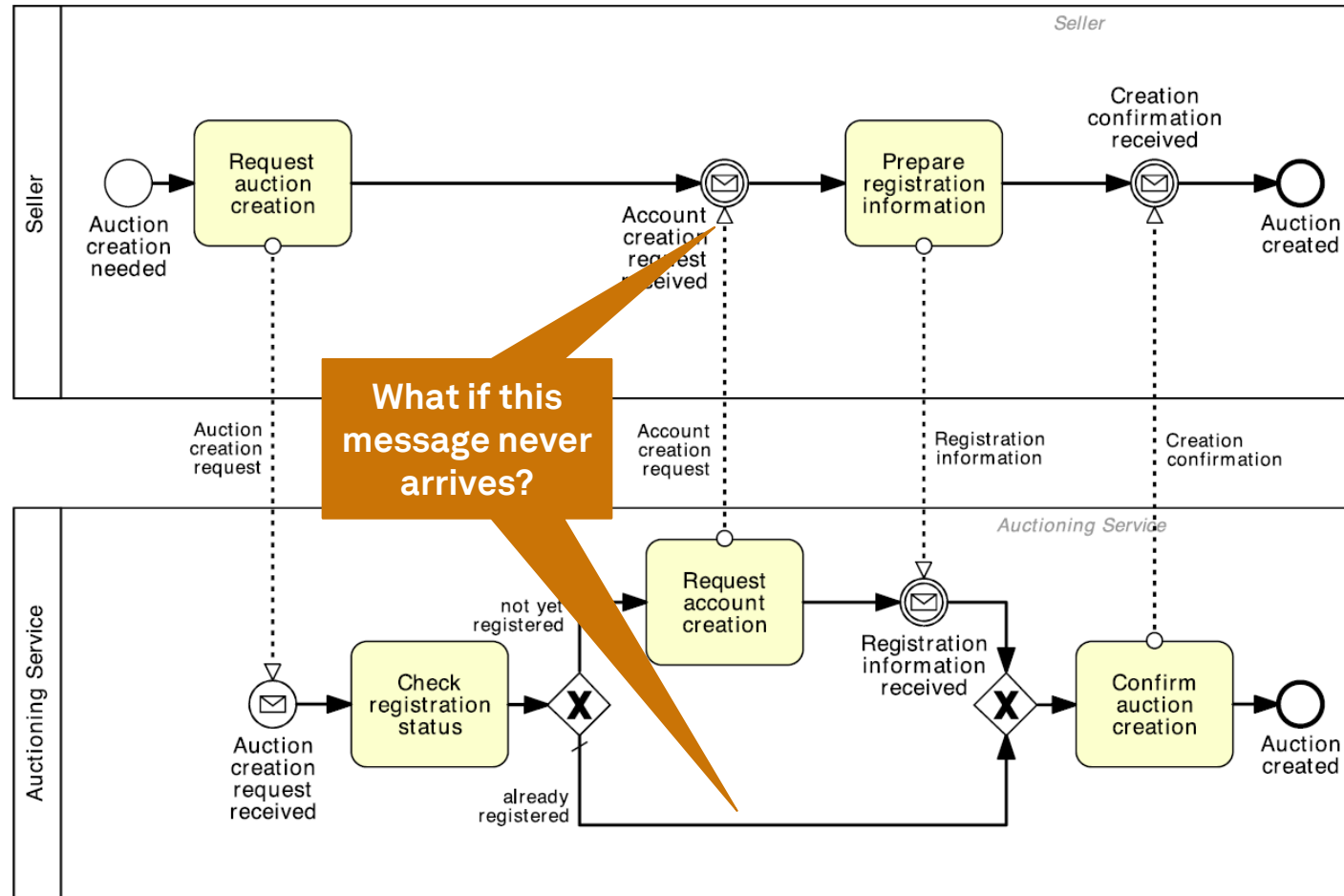
Compensation Events and Activities



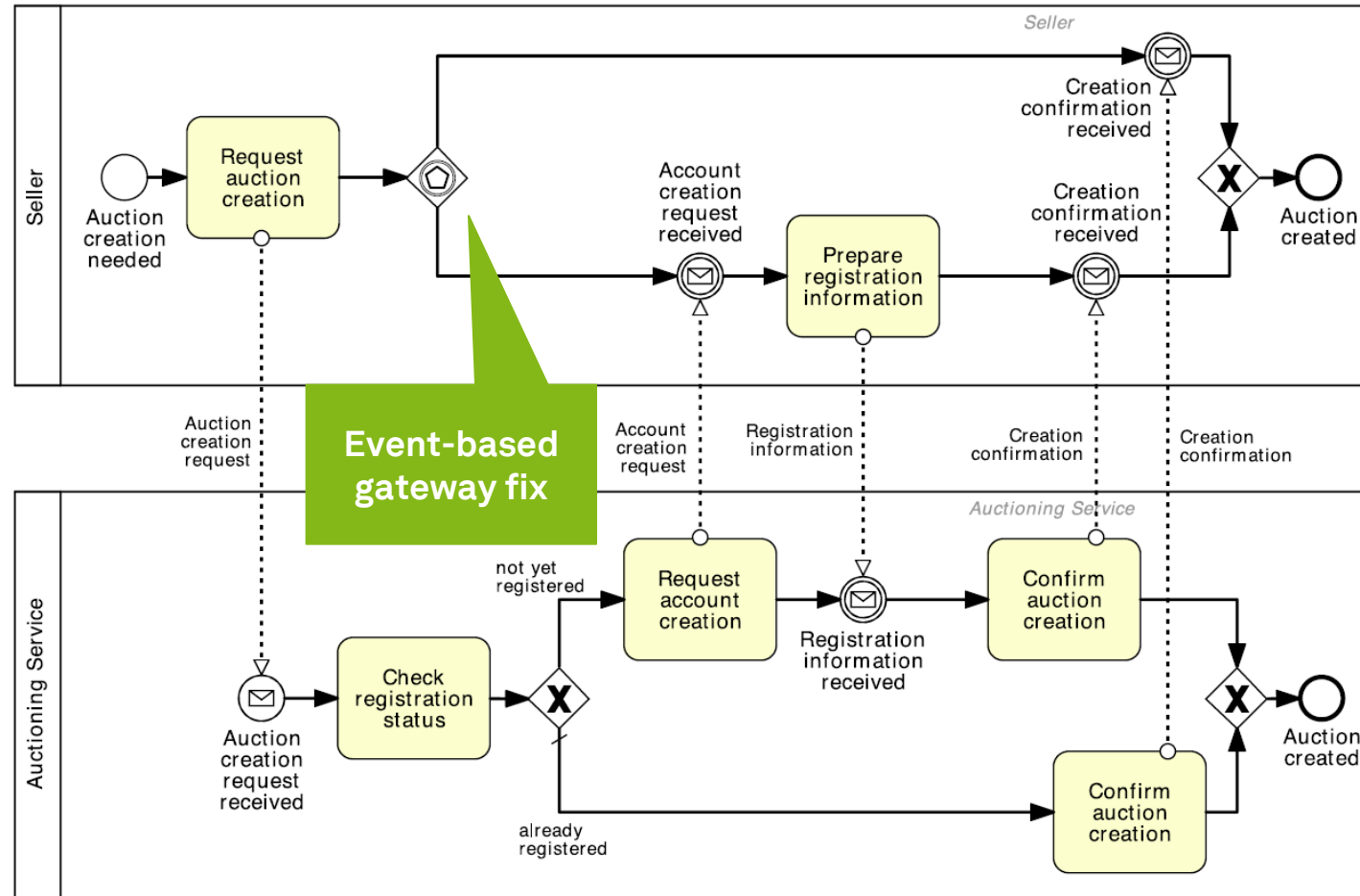
Complex Exception Handling Scenario



Variants and Collaborations: What's Wrong Here?

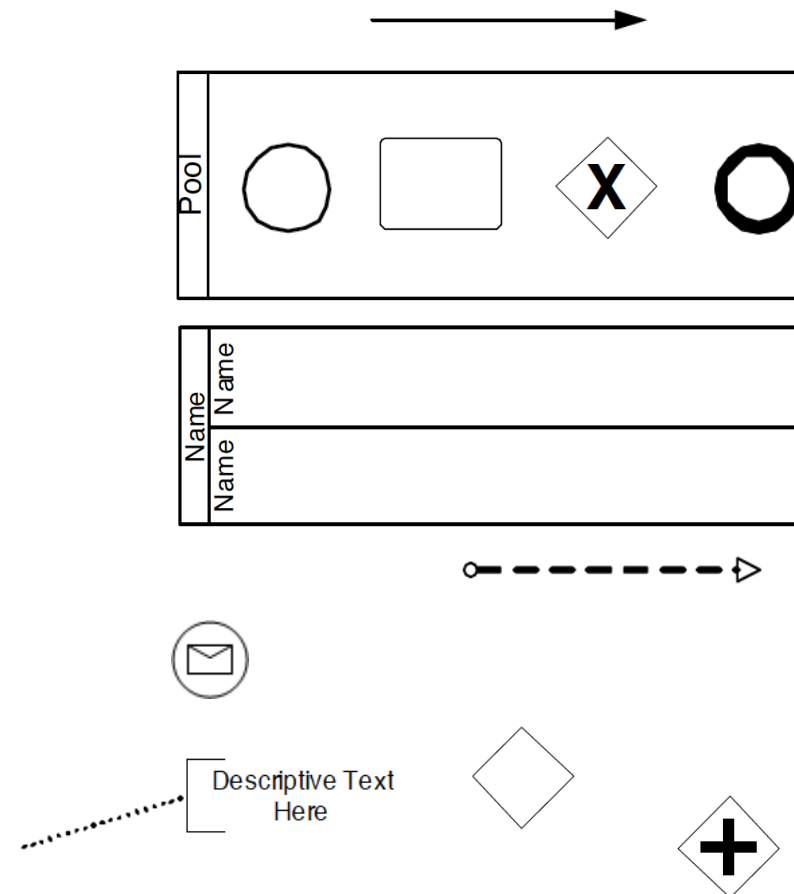
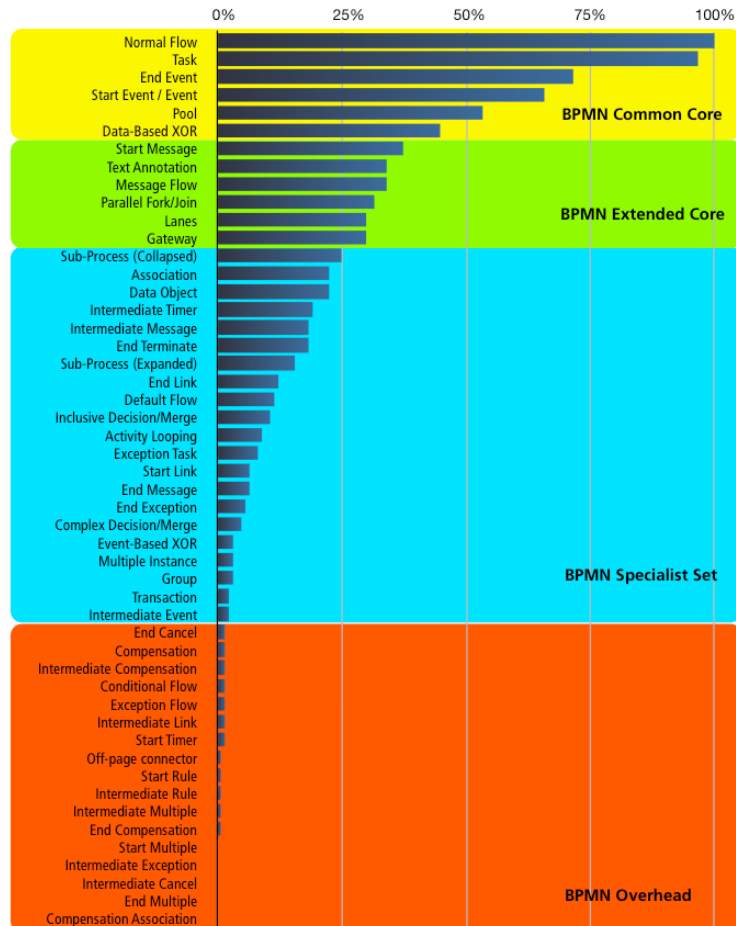


Variants and Collaborations: What's Wrong Here?



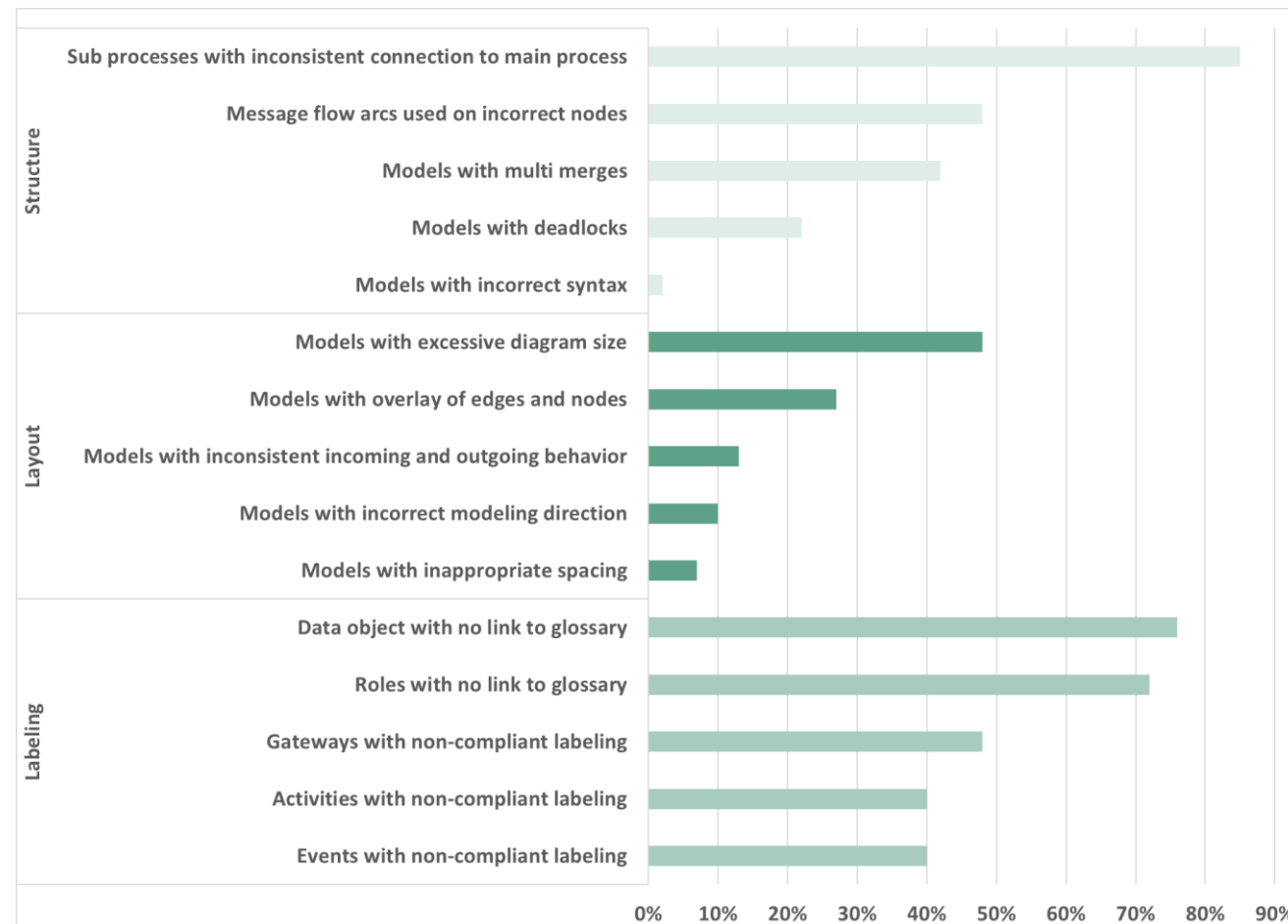
Dumas et al. (2018)

BPMN has over 100 symbols, but...



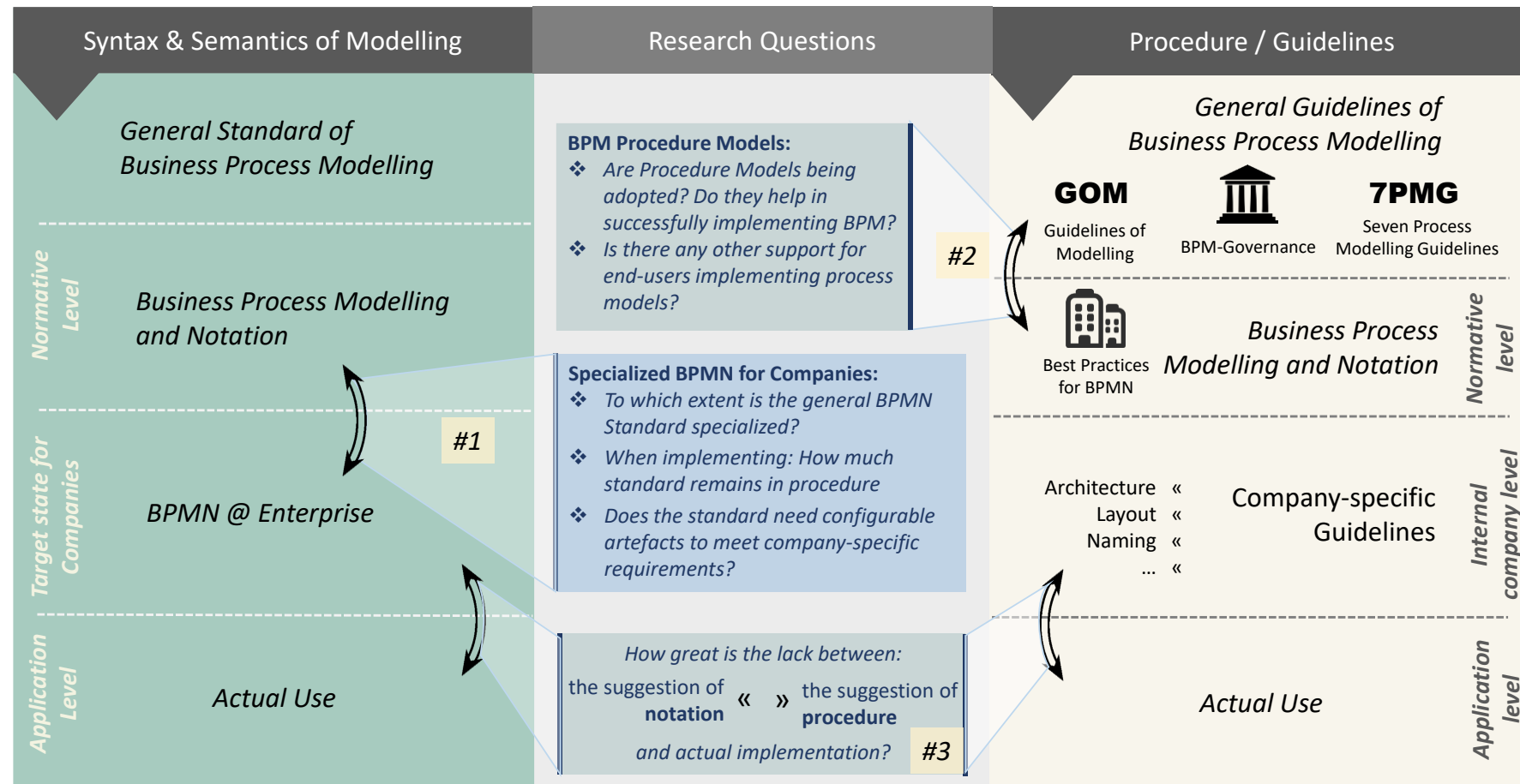
zur Mühlen, Recker (2008)

Quality Issues of BPMN Models



Derived from Leopold (2015)

BPMN Use – Levels of Abstraction



Imgrund (2015)

Workflow Patterns

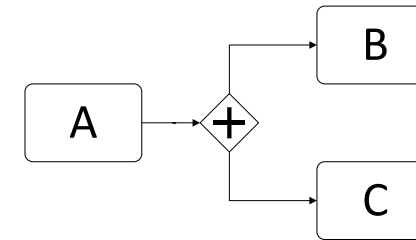


Control Flow Patterns

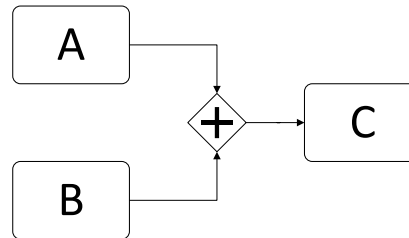
1. Sequence



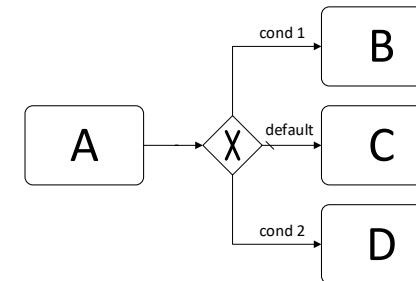
2. Parallel Split



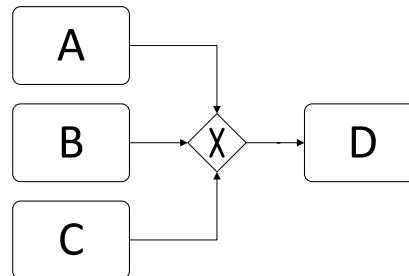
3. Synchronization



4. Exclusive Choice

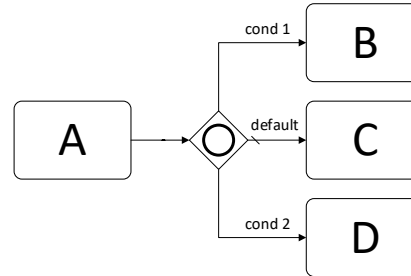


5. Simple Merge

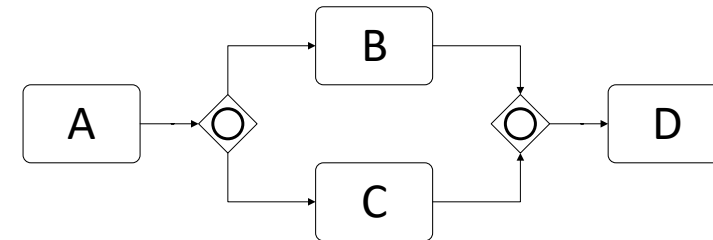


Advanced Branching and Synchronization Patterns

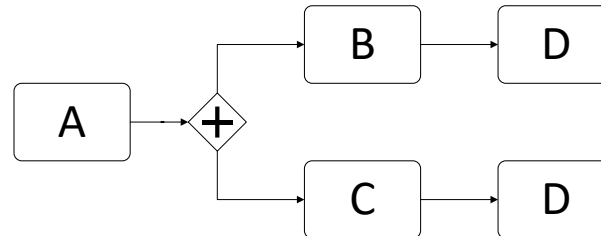
6. Multi-Choice



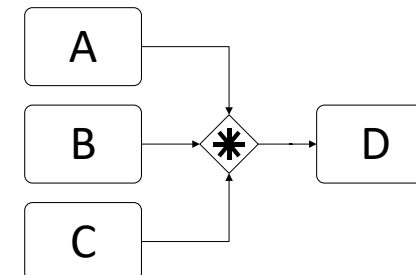
7. Structured Synchronizing Merge



8. Multi-Merge

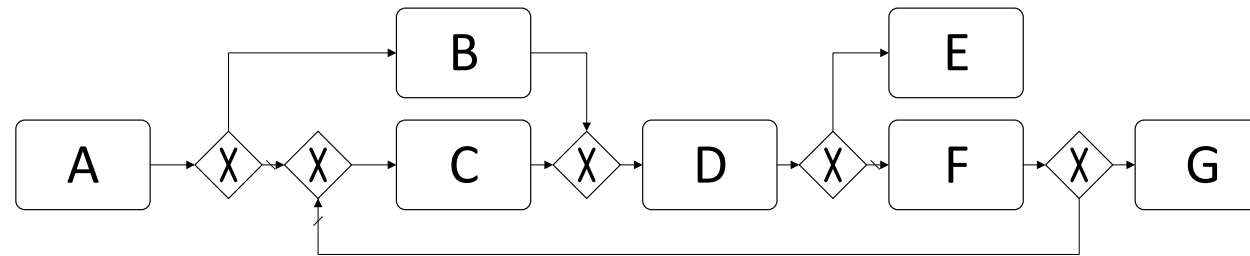


9. Structured Discriminator

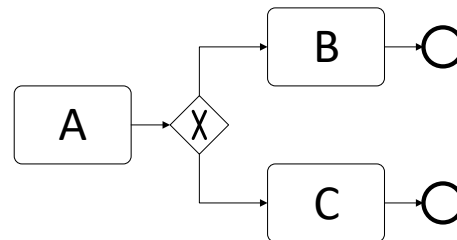


Iteration Patterns and Termination Patterns

10. Arbitrary Cycles



11. Implicit Termination

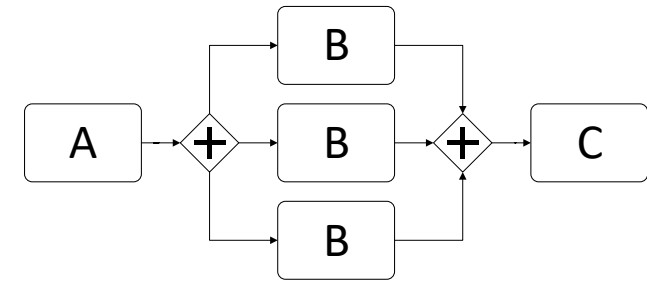


Multiple Instance Patterns

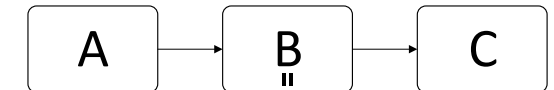
12. Multiple Instances without Synchronization



13. Multiple Instances with a Priori Design-Time Knowledge



14. Multiple Instances with a Priori Run-Time Knowledge

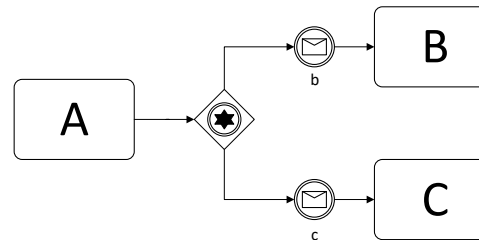


15. Multiple Instances without a Priori Run-Time Knowledge

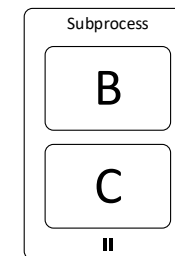
No support in BPMN,
workaround loop activity
and event sub-processes

State-based Patterns

16. Deferred Choice



17. Interleaved Parallel Routing

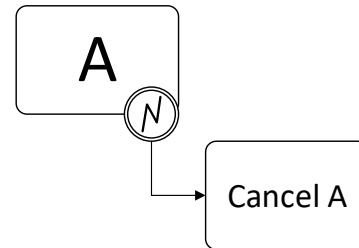


18. Milestone

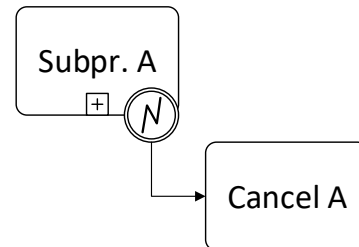
No support in
BPMN

Cancellation and Force Completion Patterns

19. Cancel Task



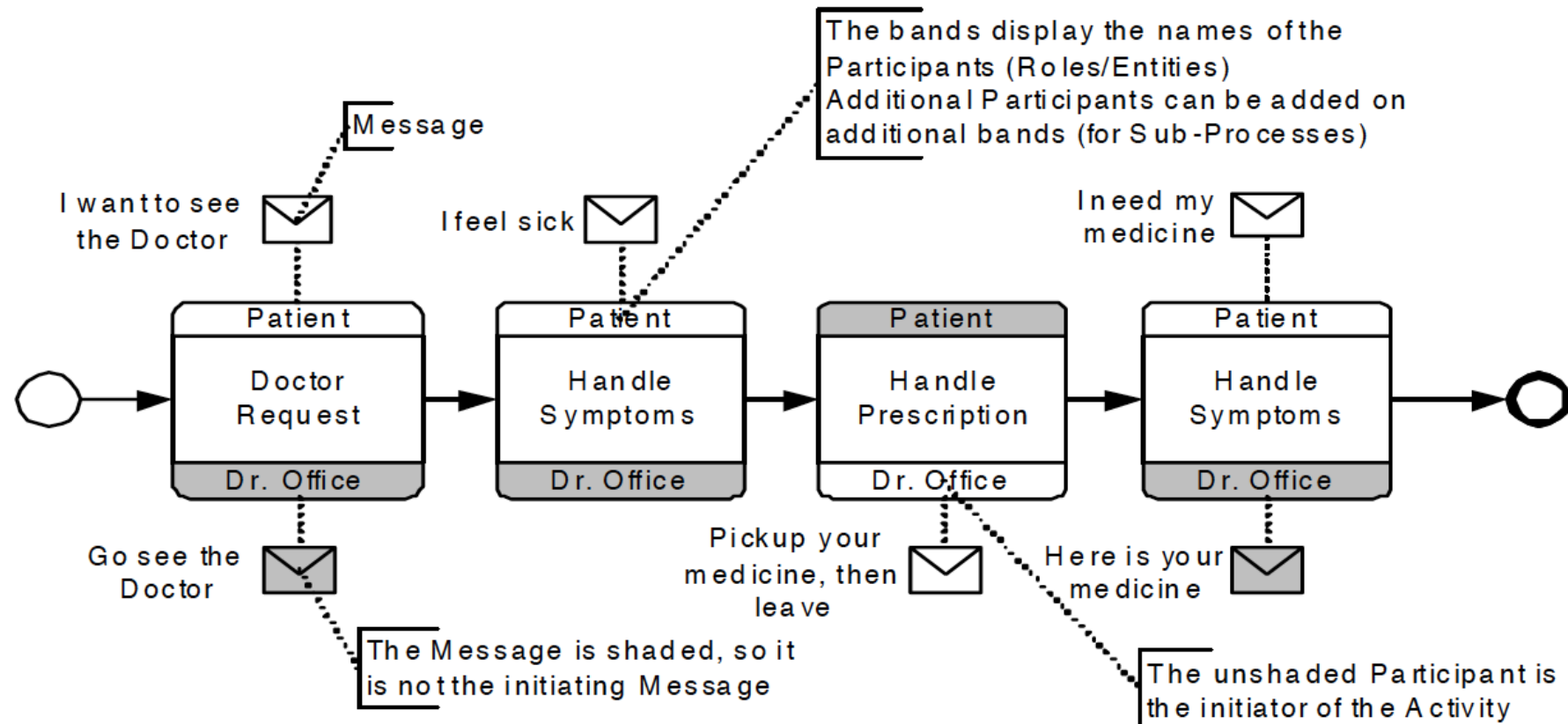
20. Cancel Case



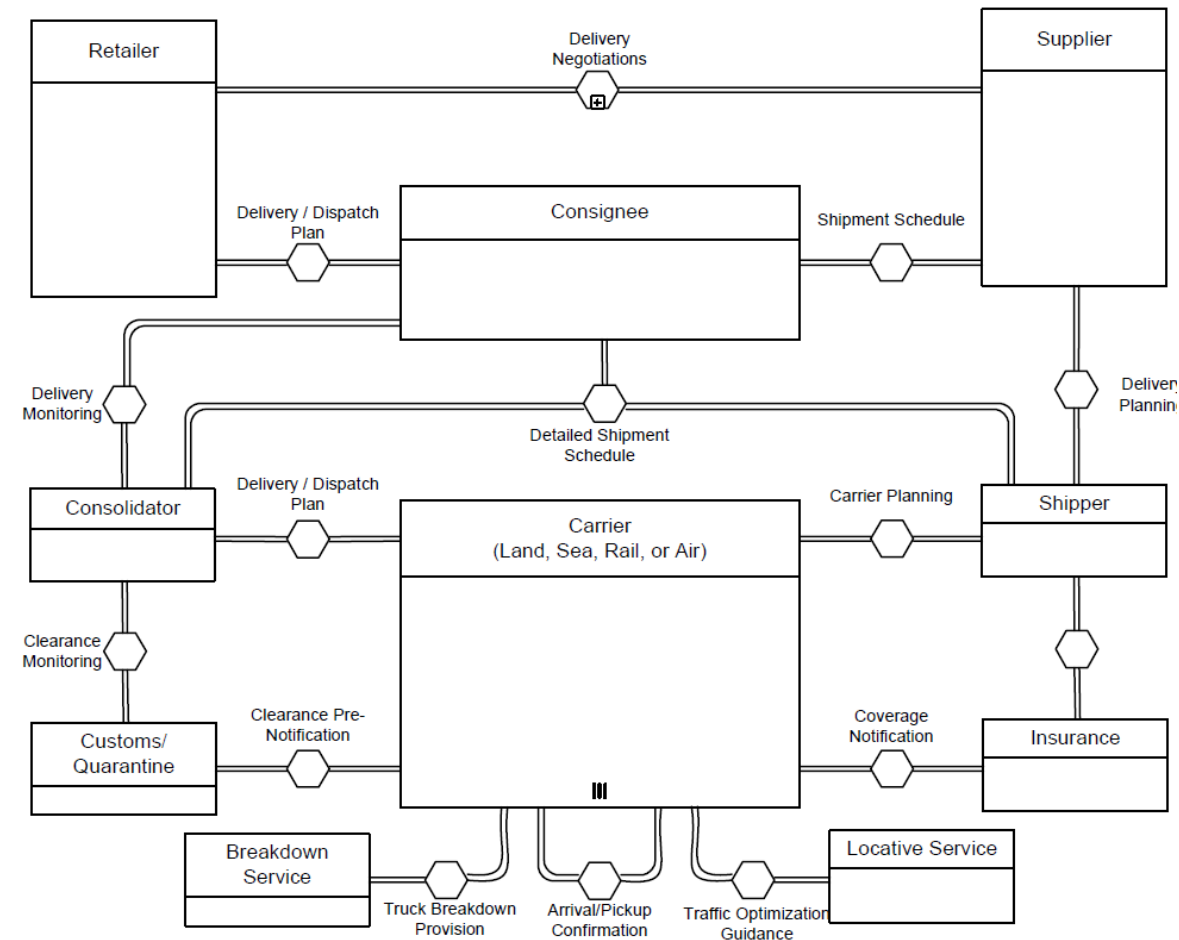
Even More BPMN!



BPMN Choreography Example

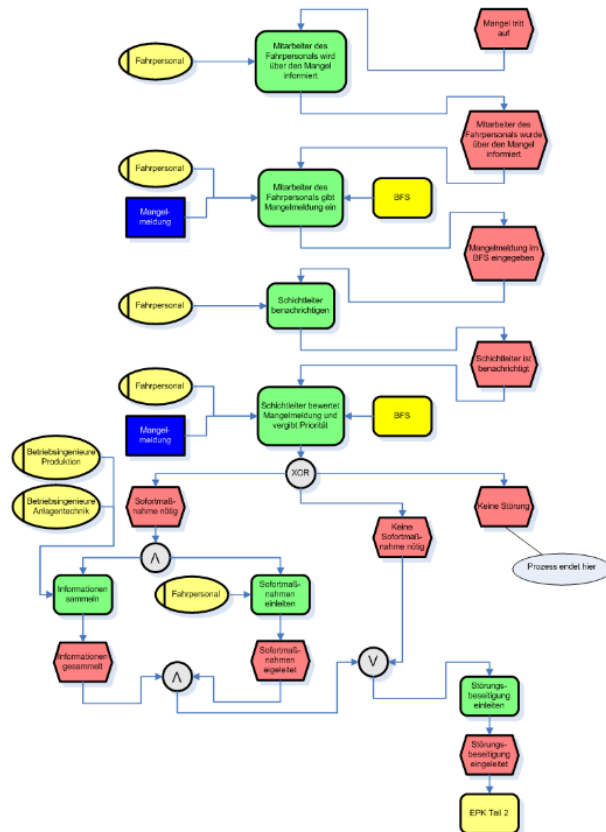


BPMN Conversation Example

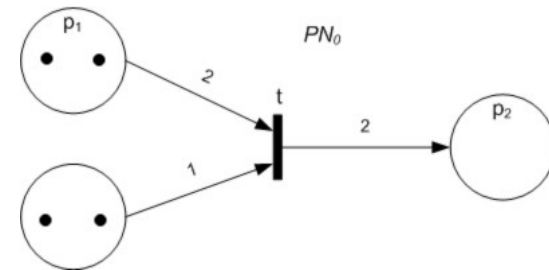


Good to know: Alternatives to BPMN

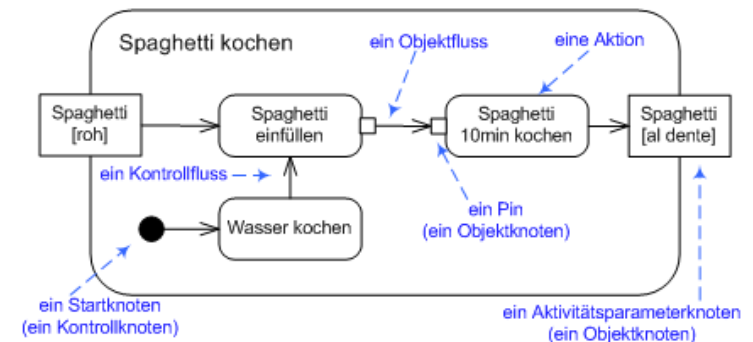
- Event-driven Process Chains



- Petri Nets



- UML 2.0 Activity Diagrams



Learning Goals of Today

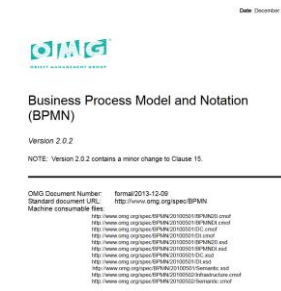
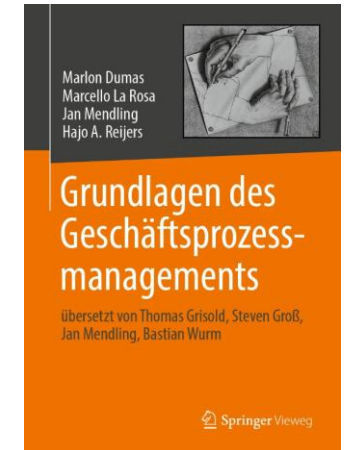
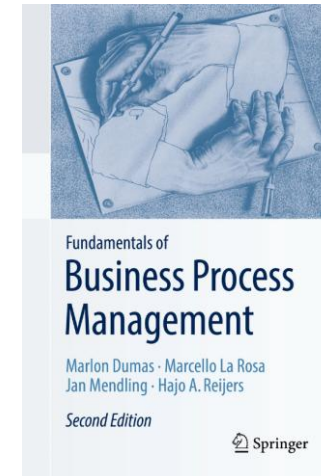
- Understand throwing and catching events in BPMN
- Understand exception handling in BPMN
- Understand and know workflow patterns
- Be able to use basically all of BPMN

Recap Questions

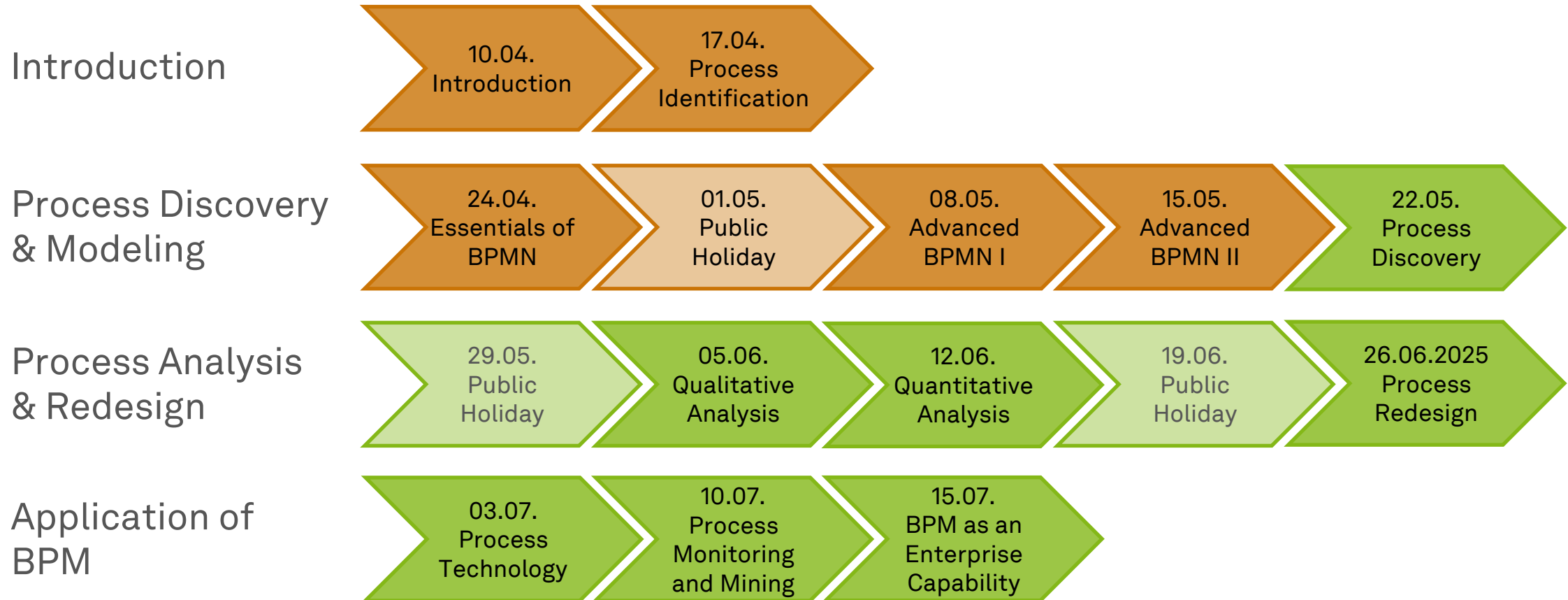
- What are racing events in BPMN and how do we model them?
- What are boundary events in BPMN and how do we model them?
- How do I model internal and external exceptions in BPMN and what is a compensation event and how do I use it in BPMN?
- What is a workflow pattern?

Literature

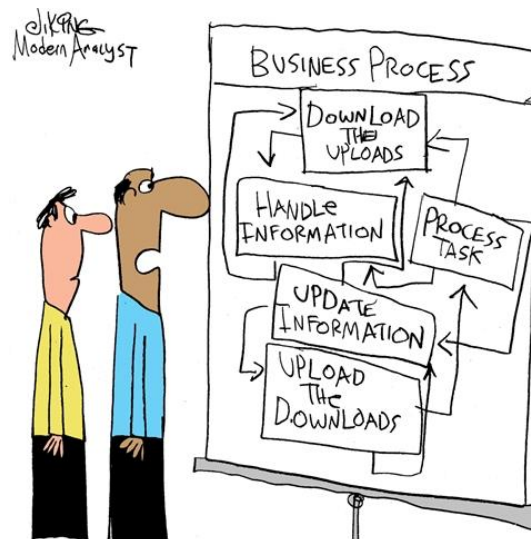
- **Section 4: Advanced Process Modeling**
- Business Process Model and Notation (BPMN)
 - <https://www.omg.org/spec/BPMN/2.0.2>



Roadmap



Any Questions about Process Modeling?



*"Charlie, did you really get these details
from the customer?"*



ec.cs.tu-dortmund.de

